3. $E^m\left[E^n f(x)\right]=E^n\left(E^m f(x)\right)=E^{m+n} f(x)$

4. If '$n$' is a positive integer, then $E^n\left[E^{-n}\left(f(x)\right)\right]=f(x)$.

**Note**

Let $y=f(x)$ be given function of $x$. Let $y_0, y_1, y_2, \cdots, y_n$ be the values of $y$ at $x = x_0, x_1, x_2, ..., x_n$. Then $E$ can also be defined as

$Ey_0 = y_1$, $Ey_1 = y_2, ..., Ey_{n-1} = y_n$

$E\left[Ey_0\right]=E\left(y_1\right)=y_2$ and

In general $E^n y_0 = y_n$

**Relations between the operators $\Delta$, $\nabla$ and E:**

1. $\Delta \equiv E-1$

**Proof:** From the definition of $\Delta$ we know that

$\Delta f(x) = f(x+h) - f(x)$

and $E[f(x)] = f(x+h)$

where h is the interval of difference.

$$\Delta f(x) = f(x+h) - f(x)$$

$$\Delta f(x) = Ef(x) - f(x)$$

$$\Rightarrow \Delta f(x) = (E-1)f(x)$$

$$\Delta \equiv E-1$$

$$\therefore \quad E \equiv 1+\Delta$$

2. $E\Delta \equiv \Delta E$

**Proof:**

$$\begin{aligned} E\left(\Delta f(x)\right) &= E\left[f(x+h)-f(x)\right] \\ &= E f(x+h) - E f(x) \\ &= f(x+2h) - f(x+h) \\ &= \Delta f(x+h) \\ &= \Delta E f(x) \end{aligned}$$

$\therefore E\Delta \equiv \Delta E$

3. $\nabla \equiv \dfrac{E-1}{E}$

**Proof:**

$$\begin{aligned} \nabla f(x) &= f(x) - f(x-h) \\ &= f(x) - E^{-1} f(x) \\ &= (1-E^{-1}) f(x) \\ \Rightarrow \nabla &\equiv 1-E^{-1} \\ \text{i.e., } \nabla &\equiv 1-\frac{1}{E} \\ \text{Hence, } \nabla &\equiv \frac{E-1}{E} \end{aligned}$$

DO YOU KNOW?

(i) $(1+\Delta)(1-\nabla)=1$

(ii) $\Delta\nabla \equiv \Delta - \nabla$

(iii) $\nabla \equiv E^{-1}\Delta$

**Example 5.1**

Construct a forward difference table for the following data

| $x$ | 0 | 10 | 20 | 30 |
|---|---|---|---|---|
| $y$ | 0 | 0.174 | 0.347 | 0.518 |

***Solution:***

The Forward difference table is given below:

| $x$ | $y$ | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ |
|---|---|---|---|---|
| 0 | 0 | | | |
| | | 0.174 | | |
| 10 | 0.174 | | −0.001 | |
| | | 0.173 | | −0.001 |
| 20 | 0.347 | | −0.002 | |
| | | 0.171 | | |
| 30 | 0.518 | | | |

**Example 5.2**

Construct a forward difference table for $y = f(x) = x^3 + 2x + 1$ for $x = 1,2,3,4,5$.



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